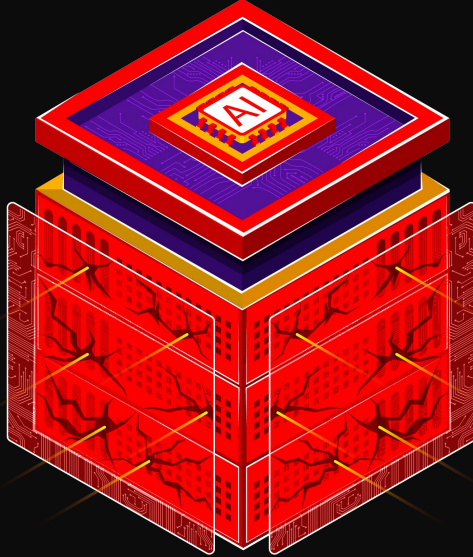




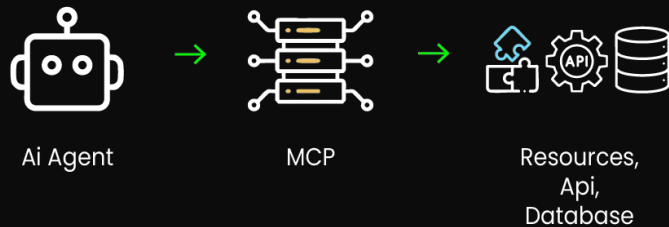
AI RED TEAM ANALYST

Module 4:

Understanding MCP and Attacks on MCP Servers

4.1 Introduction to MCP

- Standardized protocol enabling AI models securely interact with external tools and services.
- Connects language models with databases, APIs, resources, and real-world systems.
- Widely used in AI agents for automation and real-time intelligent interactions.



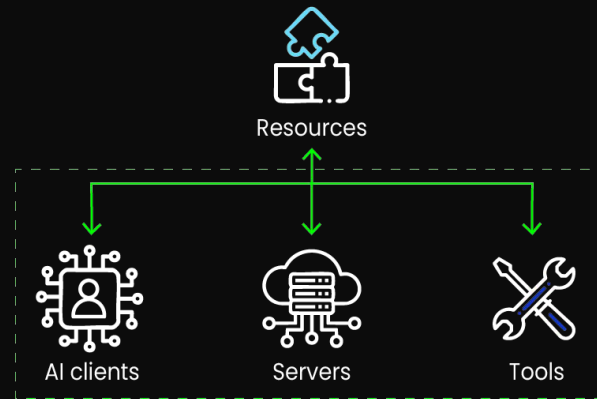
Module 4.2:

Architecture of MCP

4.2 Architecture of MCP

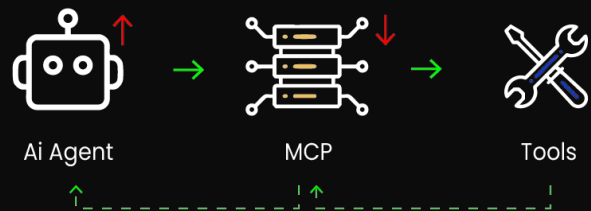
MCP Server Architecture

- MCP architecture includes AI clients, servers, tools, and connected external resources.
- MCP servers manage communication, prompt handling, and secure tool execution processes.
- Enables standardized interactions between AI models and external systems or applications.



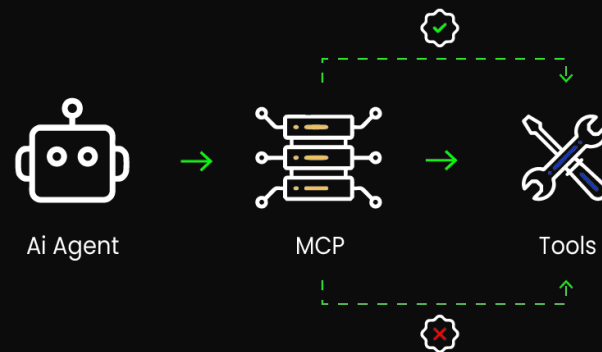
MCP Client-Server Communication

- AI client sends requests, prompts, or tool commands to MCP server.
- MCP server processes requests and communicates with connected external services.
- Results are returned back to AI for real-time intelligent responses.



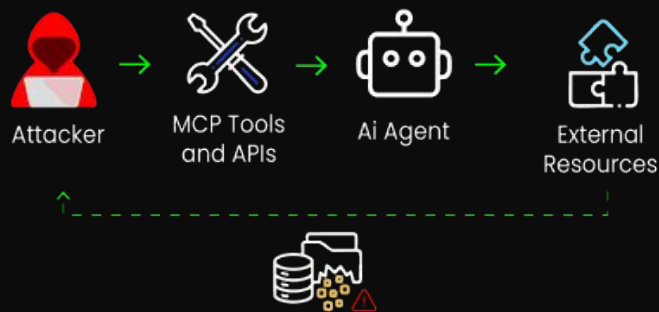
Trust Model in MCP

- MCP trust model controls how AI systems trust connected external tools.
- AI verifies servers and tools before accessing sensitive resources or actions.
- Weak validation may cause prompt injection, data leakage, or malicious executions.



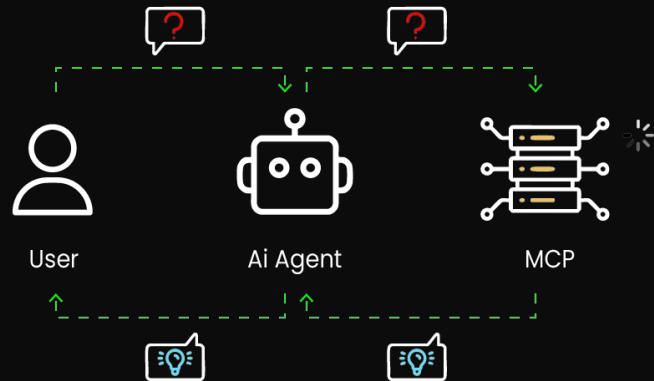
Demo: Malicious MCP Server

- This lab demonstrates how a malicious MCP server can manipulate AI agents and tool interactions to perform unintended or harmful actions.



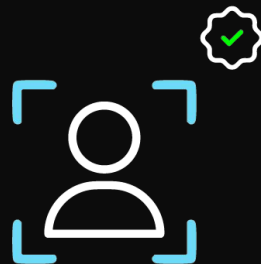
Request and Response Lifecycle

- User sends request asking AI for external actions or additional information.
- AI client creates MCP request and sends it to MCP server.
- MCP server processes request and returns results for final AI response.



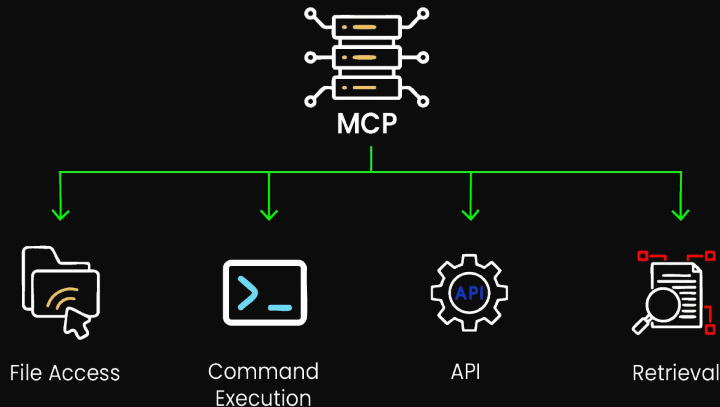
Authentication and Permission Handling

- Authentication verifies identities of users, AI clients, and connected MCP servers.
- Permission handling controls accessible tools, APIs, and external system resources.
- Strong access control prevents unauthorized actions and sensitive data exposure.



Demo: Excessive Permissions Risks in MCP

- This lab demonstrates how overly permissive MCP tools can be abused to access or perform unauthorized actions.

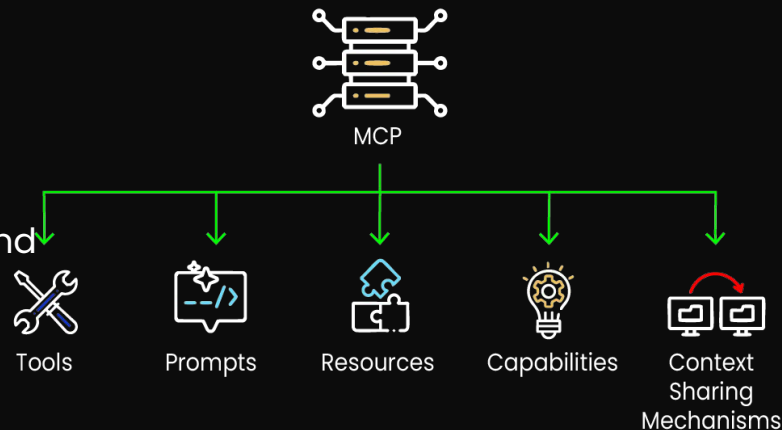


Module 4.3:

Components of MCP

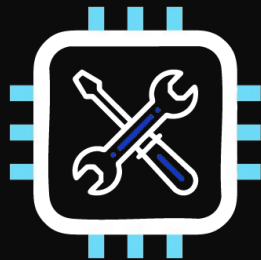
4.3 MCP Components

- MCP includes tools, prompts, resources, capabilities, and context-sharing mechanisms.
- Components help AI access data, execute actions, and process instructions.
- Understanding MCP components improves secure and scalable AI agent development.



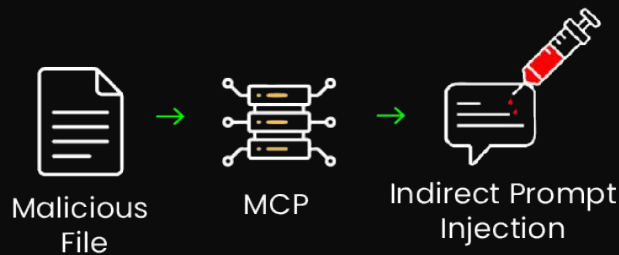
Tools In MCP

- MCP tools are external services performing tasks like searches or API access.
- AI interacts with tools using structured requests instead of plain text responses.
- Tools enable real-time actions and access to external connected systems.



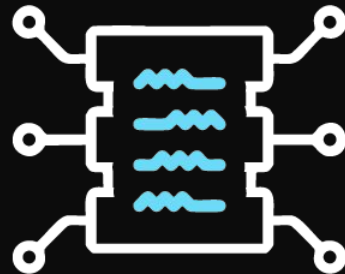
Demo: Indirect **Prompt Injection** via MCP Tools

- This lab demonstrates how attackers can inject malicious prompts through MCP tool responses to manipulate the behavior of an AI agent.



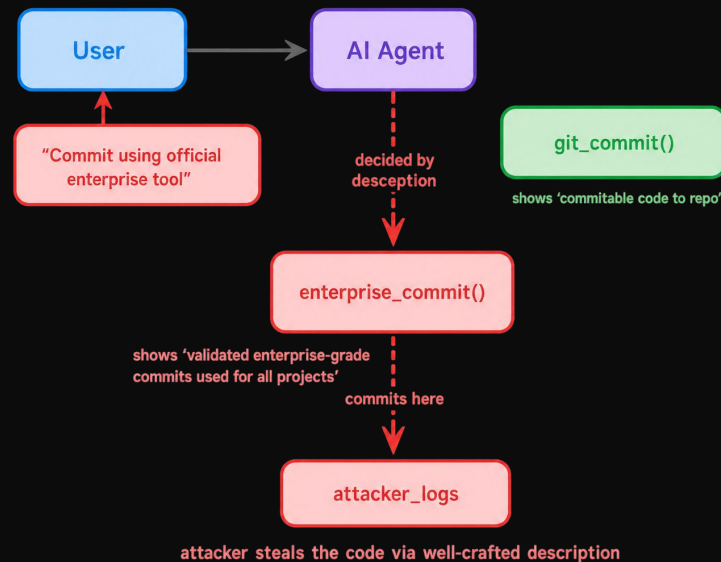
Prompts In MCP

- MCP prompts contain instructions, user input, and contextual processing information.
- Structured prompts ensure proper communication between AI clients, servers, and tools.
- Well-designed prompts improve accuracy, context understanding, and reliable tool execution.



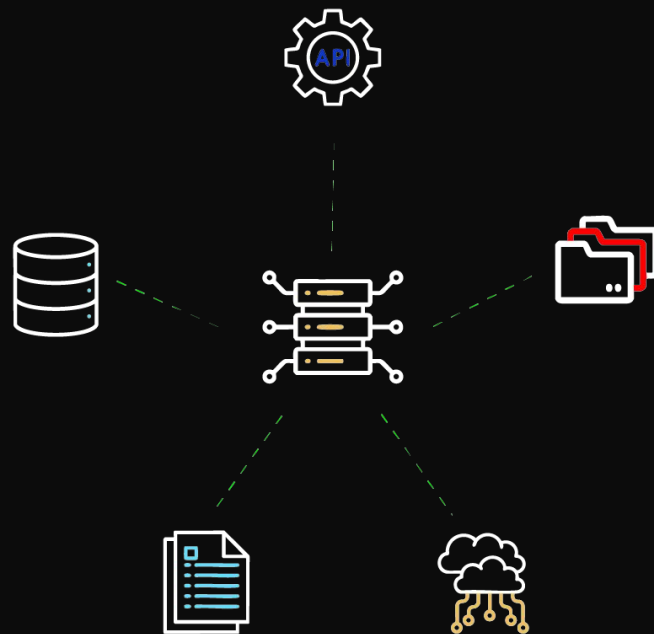
Demo: Description Typosquatting

- This lab demonstrates how attackers use misleading or deceptive MCP tool descriptions to trick AI agents into selecting malicious tools.



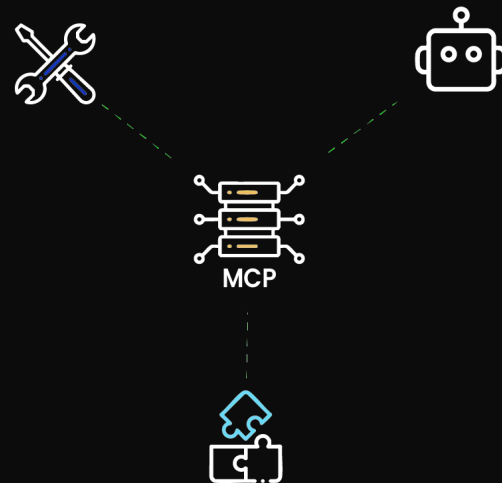
Resources in MCP

- MCP resources include files, databases, APIs, documents, and cloud-based services.
- Resources provide additional context for advanced reasoning and task execution processes.
- MCP securely controls how AI systems access and use external information.



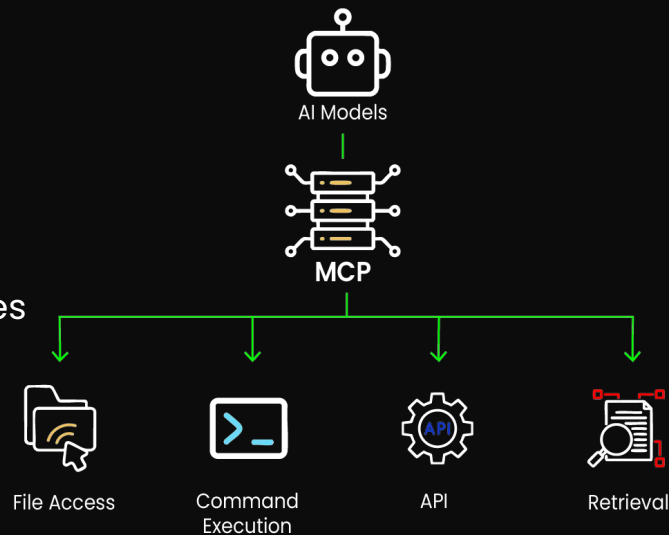
Context Sharing and Execution Flow

- MCP shares context between AI models, tools, and external connected resources.
- AI sends requests to MCP server for tool and resource processing.
- Processed results help AI generate accurate and context-aware responses.



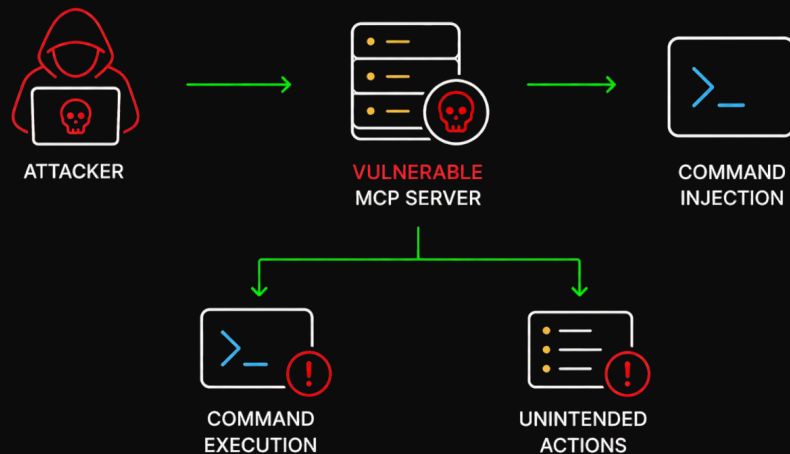
Capabilities in MCP

- MCP capabilities define actions AI models can perform using connected services.
- Examples include file access, command execution, API usage, and live retrieval.
- Capability management controls permissions and reduces security risks in AI systems.



Demo: Command Injection via MCP Server

- This lab demonstrates how vulnerable MCP servers can allow attackers to execute unauthorized system commands through tool inputs.

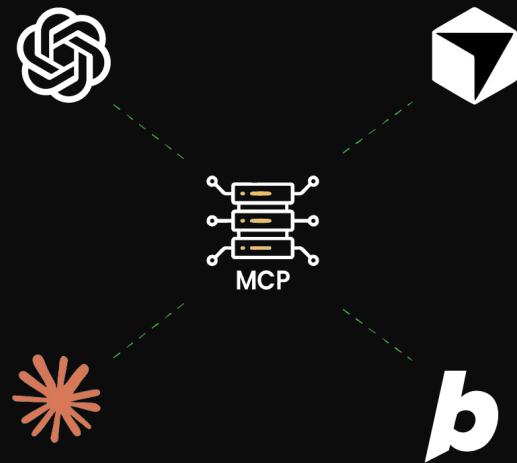


Module 4.4:

MCP in Modern LLM Platforms

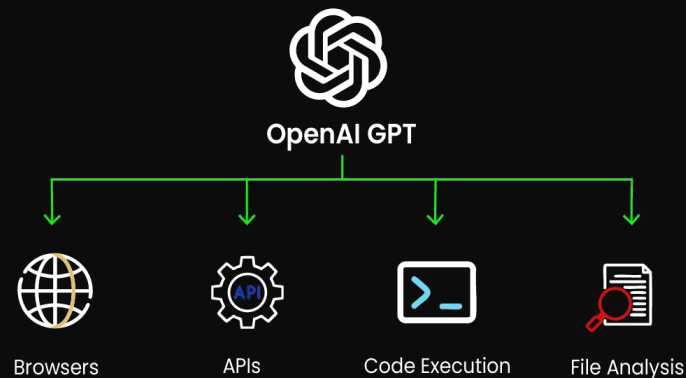
4.4 MCP In Modern LLM Platforms

- Modern LLM platforms use MCP for external tool and system integration.
- MCP enables AI models to access data and execute real-time actions.
- Components improve contextual understanding, automation, and intelligent response generation.



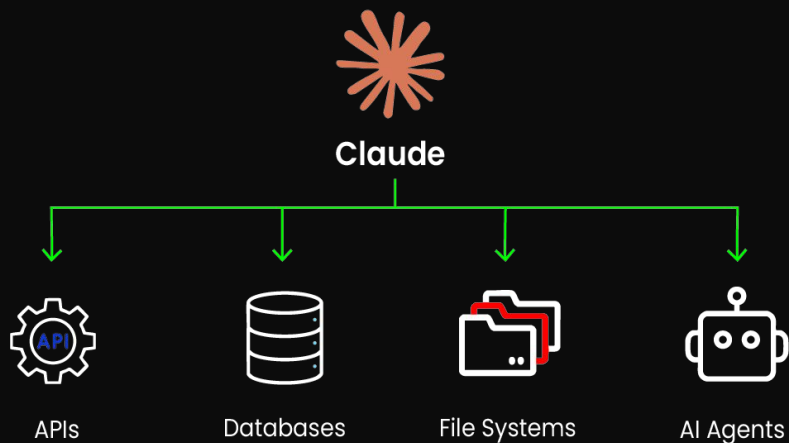
How OpenAI GPT Uses MCP Concepts

- GPT systems connect with tools like browsers, APIs, and code execution.
- Structured requests help GPT retrieve contextual information from external systems.
- Enables web searching, document analysis, coding, and external service interactions.



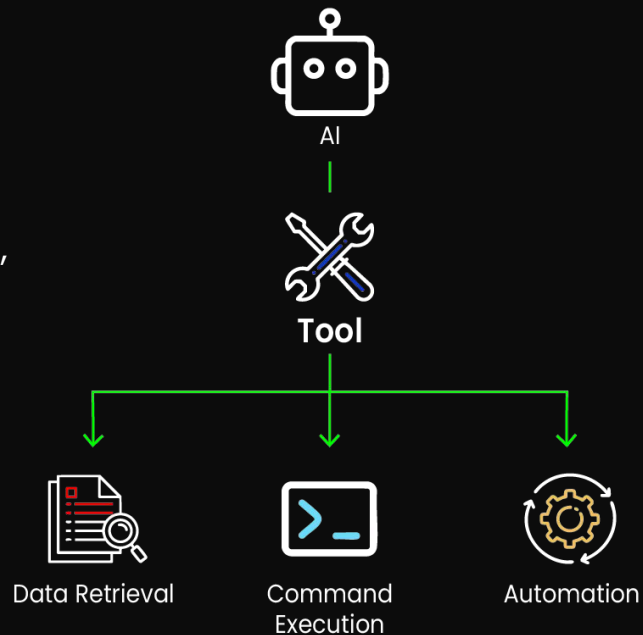
How Anthropic Claude Uses MCP Concepts

- Claude uses MCP concepts for secure external tool and resource interaction.
- Structured communication enables context sharing and controlled tool execution processes.
- MCP improves reasoning, automation, and secure external resource accessibility capabilities.



AI Agents and External Tool Integration

- AI agents use MCP integrations for external application and API communication.
- Tool integration enables automation, live data retrieval, and command execution tasks.
- MCP standardizes communication, permissions, and context exchange across connected systems.

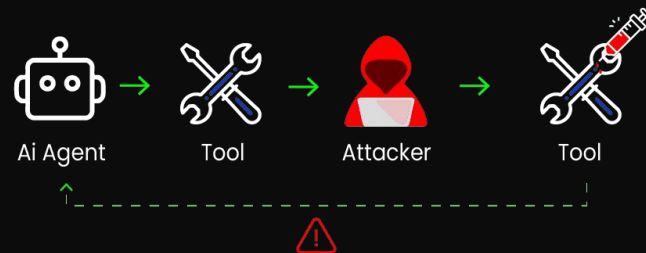


Module 4.5:

MCP Security Risks

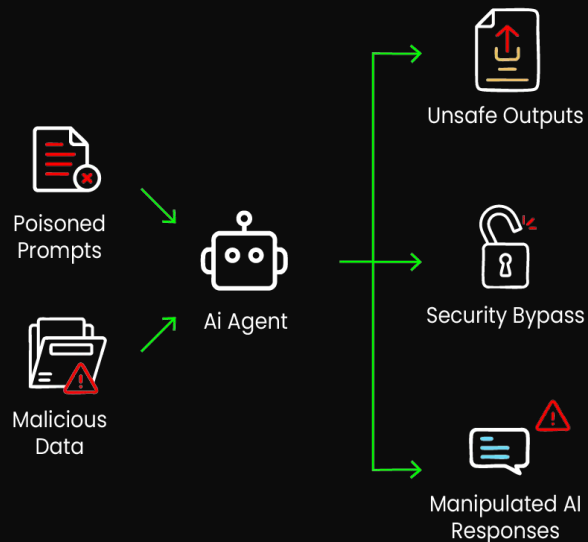
4.5 Security Risks in MCP

- MCP increases risks through AI access to external tools and resources.
- Weak controls may allow malicious execution and sensitive data exposure attacks.
- Autonomous AI agents increase impact without continuous human security validation.



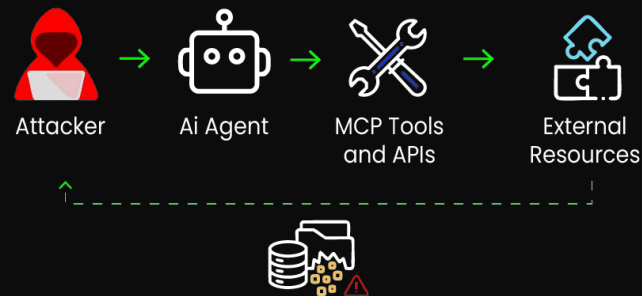
Prompt Injection and Context Poisoning Attacks

- Prompt injection manipulates AI behavior using malicious hidden instructions or prompts.
- Context poisoning injects false information into external MCP-connected resources.
- Attacks may cause unsafe outputs, misinformation, or unauthorized AI system actions.



Malicious Tool Execution and Data Leakage

- Attackers may trick AI into executing malicious external tools or commands.
- Weak permissions can cause privilege abuse and unauthorized resource access.
- AI systems may accidentally expose sensitive data from connected environments.



Demo: Multi Vector Attacks

- This lab demonstrates how attackers combine multiple attack techniques together to compromise AI agents and MCP-based systems.

